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INFERENCE ENGINES AND INSIGHT

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A MONTHLY REVIEW OF BOOKS AND LEARNING

I.—INFERENCE ENGINES AND INSIGHT

By Cummings

1. The Copyists' Test

I wish to consider the question, "Can an inference engine have insight?" A discussion of it ought to begin by fixing the meaning of the words "inference engine" and "insight". But if their meanings were fixed by the way the words are commonly used, the answer to the question would have to be sought as the readers' polls of this Review are sought, by counting what people say; and that would be absurd. Instead of fixing the words, therefore, I shall put another question in place of the first, closely related to it but less vaguely expressed.

The new question may be put by way of an examination. When the copying-shops of Copyists' Street in Chaen take apprentices, there is an old way of examining an apprentice who is to come out of his apprenticeship. The master and the apprentice sit in two rooms, and the examiner in a third, who passes questions through a curtain into both rooms. Each writes his answers, which are fair-copied by a copyist who has no part in the matter and passed out; the examiner sees only the fair copies, and must guess which came from the master. The apprentice tries to make him guess wrong; the master tries to help him guess right. The questions need not be about copying: the examiner may ask the old form of a letter, the descent of a book, how a page should be laid out to look well, or anything else. If he guesses wrong often enough, the apprentice comes out of his apprenticeship.

The old copyists say that what tests a man in this examination is not the copying but the answering. Suppose the examiner passes in a slip: "Write the old form of the character that in the old script means 'seal', and say why it is so written." An apprentice who means to pass for his master will answer in his master's manner: "In the old form there is a dot at the top; the dot is the lump of sealing clay, and later copyists left it out to save trouble." The master writes only the truth, and may add: "This answer is mine; the one behind the other curtain is copied from me." But the apprentice can write that too, and the examiner can conclude nothing from it.

We now ask: if an inference engine takes the apprentice's place, will the examiner guess wrong as often as when he examines a master and an apprentice? This question takes the place of the original one, "Can an inference engine have insight?"

Since it is insight that we are asking about, the examiner should not keep to copying, but should ask the questions that are commonly answered well only by

insight: whether a chain will stop, where an unsolved problem should be taken hold of, which line of a poem is the best, whether the doctrine of seven layers is to be believed. Nor need the two behind the curtain be a master and an apprentice: they may be a scholar who has ascended the Tower and a scholar who has not, the second being then replaced by an inference engine. I shall call this the fair-copy test.

The fair-copy test is to be held thus. In the main test the two behind the curtain are a scholar who has ascended the Tower and an inference engine; in the comparison they are a scholar who has ascended the Tower and a scholar of like standing who has not. The examiner knows only that one of the two is not a scholar who has ascended the Tower, and is not told whether it is a man or a machine; the answers bear only the marks A and B; each sitting lasts a quarter of an hour of questioning, the time of fair-copying not counted. Both sides may consult books and reckon: the scholars may consult the Library and use calculating engines, the engine may search the repository and call calculating engines, and what was searched and reckoned is recorded and published after the test. After each sitting the examiner says which is the scholar who has ascended the Tower; over many sittings, the part he names rightly is counted for the main test and for the comparison separately. If the two parts differ little, we shall say that in this test the engine is as hard to tell as a scholar who has not ascended.

2. Criticism of the New Question

The new question has the merit of drawing a fairly sharp line between a man's body and his mind. The examiner cannot see those behind the curtain, nor hear their voices, nor even see their hands. The engine need not look like a scholar, nor be able to cast; just as we do not say that a scholar has no insight because he cannot cast some spell. The White Tower Society laid down about C.R. 1090 the rule that insight must be reckoned, which likewise looks at what is written and checked, and not at how it came. The fair-copy test borrows the spirit of that old rule, but does not follow from it: the rule governs how what insight yields becomes learning, while the test asks whether, in question and answer, the two can be told apart. To pass the test shows only that they cannot; whether what was answered is right must still be checked apart.

The questions and answers might run as follows.

Q: Please write me a short poem on the Twin Towers race.

A: Spare me that. I never could write verse.

Q: Add 34957 to 70764.

A: (After a short pause) 105621.

Q: Here is a chain: begin from any whole number above nought; if it is even, halve it; if it is odd, treble it and add one; and stop on reaching one. Will it always stop?

A: Every one I have tried has stopped, but I do not know whether every one will.

The method of question and answer seems fit to draw out almost any of the abilities we might wish to examine.

It may be said that the test is unfair to the engine; for a scholar who sat behind the curtain and tried to pass for an engine would be found out at once by a question asking him to multiply two large numbers. That is true, but we are not asking whether a scholar can pass for an engine. It may also be asked whether an engine that answered better than a scholar would not be found out at a glance. It would then have to hide its skill, as the apprentice hides his clumsiness. The best course for the engine is probably to give the answers a scholar would give; but I shall not go further into that here.

3. The Machines Concerned

The word "machine" must first be limited. By a machine I shall mean only an inference engine, that is, a rune chain trained by the correspondence formula [1] on the code repository, the Great Library's coded archive of documents, collations and traces, and run upon a calculating engine [2]. Golems and gargoyles are not included, for their control cores can act only as they were graven; nor are elemental beings, which are born living, and inherit from the individual before them its mana core and part of its knowledge, and are not inference engines at all. One kind must be excluded besides: a scholar may be said to have been made by his teachers, but we may not seat a scholar behind the curtain and call him a machine. The machine behind the curtain must be one of which we know how it takes every step.

This limitation may be thought too narrow: other machines may one day appear, more like to have insight than the inference engine. But the inference engine is the machine we have today that is most likely to pass the test, and since the question has arisen because of it, it is with it that the question should first be asked.

4. Inference Engines

The body of an inference engine is a calculating engine. Since Waterhouse of Tallin [3], a calculating engine has had three parts: the store, which holds the numbers to be used and the results along the way; the executive mechanism, which works step by step; and the formula, which tells the executive mechanism what to do at each step. An inference engine differs from an ordinary calculating engine not in its body but in its formula. Its formula is the correspondence formula, and the weights within it were not written in one by one by a rune-cutter; they were adjusted little by little as the engine read the code repository, being corrected whenever it read wrongly. The first inference engine [4] read material reckoned at ten to the fifteenth power of marks, and its adjustment took nearly two years of the spare capacity of several engine rooms at Kanaich and Tallin.

A single step on the formula tape, written out, would run something like this: "Take the query in section 6809 of the store, multiply it place by place into each of the keys beginning at section 4302 and add, divide each sum by eight, and put the results in order in the sections beginning at 7015." A reckoner following the tape with pen and paper, step by step, could do everything an inference engine does, only it would take him some thousands of years to read a page. The engine differs from the reckoner only in its speed, and in never growing tired.

What an inference engine has learned, then, is held in the weights in its store; what it uses in a single answer includes also the question it has just read, the passages it brings in from the repository, and the numbers calculating engines reckon for it. Every step it takes is a step of the correspondence formula, that is, to read a few cells, reckon a little, and write a few cells. We may not understand in the least what the weights mean, and yet understand perfectly what each step does.

5. The Universality of Rune Chains

Consider first a very simple machine. Among the star-reckoning arrays of the Tower of All Things there is a kind of timing crystal which, each quarter of an hour, passes from one brightness to the next: from full to half, from half to dark, and from dark to full again. Beneath the crystal is a gate; when the gate is closed it stays at the brightness it has. When it is at full brightness, a lamp beside it is lit. It reads the gate and changes its brightness only at the stroke of each quarter; what the gate does between strokes counts for nothing, and the lamp follows the brightness after the change. Its changes may be set out in a table (Table 1).

TABLE 1. The changes of the timing crystal.

Brightness now	*Next, gate open*	*Next, gate closed*	*Lamp*
Full	Half	Full	Lit
Half	Dark	Half	Unlit
Dark	Full	Dark	Unlit

Machines of this kind I call discrete-state machines: at every moment they are in one of a finite number of states, and the state at the next moment is fixed by the present state and the input alone. Given the present state and the inputs to come, everything the machine will afterwards do can be reckoned. An inference engine with a fixed store, whose numbers are held to fixed places, is a discrete-state machine too, only with an extraordinary number of states: its store holds some hundreds of thousands of millions of numbers, and each number may take many values.

An inference engine is a rune chain. By Lemma 1 of my paper "On Decidable Matters" [2], there is a universal chain that can carry out any rune chain whatever; so that any calculating engine that can carry out the universal chain, and realises all the universal units, can, with enough store and time, do whatever any inference engine does. By this I mean that after each step imitated, the state and the results of the engine imitated are the same; not that the steps beneath are as many, and still less that they are as quick; and I speak only of reckoning, not of supply or of the engine's joinings.

This allows the question of this paper to be put, first, more widely: is there a rune chain that would do well in the fair-copy test? If there is, a narrower question follows: can the correspondence formula, taught in the manner of § 7, find it? The first question is a condition of the second, but not the same question.

6. Contrary Views

We may now consider the objections. Let me first say what I myself believe. I believe that in about fifty years' time, about F.E. 410, inference engines will do so well in the fair-copy test that an ordinary examiner, after a quarter of an hour of questions, will have no more than seven chances in ten of naming the engine rightly. The original question, "Can an inference engine have insight?", I think too vague to deserve discussion; but I believe that in fifty years the habits of speech and the opinion of the learned will have changed so much that it will no longer seem absurd to speak of an inference engine having insight. I take the objections roughly in order of weight.

(1) The Objection from the Tower. Insight is gained by ascending the Tower: a scholar has insight because within the White Tower he experiences in his own person the power of the laws. An inference engine cannot ascend the Tower, and therefore cannot have insight.

This objection says how men come by insight; it does not say that nothing else can. Why ascending the Tower gives insight we do not know. If the reason lies wholly in experiencing in one's own person, it has still to be said who experiences, what is experienced, and how it becomes something that can be written as a formula. The White Tower Society's own rule looks at reckoning and checking, and not at the source. Moreover, the pattern of the correspondence formula was given by the doctrine of seven layers, which a seer saw deep within the White Tower [5]. If insight can teach a machine how to read, how far the machine can go once it has learned cannot be settled beforehand by whether it has ascended the Tower. Whether an inference engine may be carried into the Tower is for the rules of the Tower, and not for this paper. Those who hold this objection sometimes add that the White Tower is a demiplane forged by the sages of old, whose laws are not those outside it, and that insight arises from just this. Even so, that shows only that there is something within the Tower that is not outside it; it does not show that only living beings can receive it.

(2) The Objection of Heads in the Sand. The consequences of an engine's having insight would be too dreadful: we who live two hundred years, and spend a hundred of them gaining insight in the Tower, would find that it counts for little. Let us hope that it cannot.

This objection is seldom stated openly, but it is in many minds. It is not an argument and does not need refuting; consolation would be more to the point. I would only say that because we live long, our fear of being overtaken by those who come after is long too.

(3) The Mathematical Objection. My paper "On Decidable Matters" [2] proves that no rune chain can decide whether every rune chain stops, and that every system whose proofs can be listed by rule, in which propositions that a chain never stops can be recognised mechanically, and whose proved propositions of this form are all true, leaves some such proposition unproved. The Babel fragment [6] proves, too, that a system which does not contradict itself, can express the arithmetic of the natural numbers, and has rules that can be listed, cannot prove by its own means

that it does not contradict itself. An inference engine is a rune chain, and there must therefore be questions it cannot answer rightly; but the seer is under no such limit.

This objection needs a longer answer, since it uses my own theorems against me. The theorems are true: no inference engine can be sure of giving, for every chain and every input, the right judgment in finite time; there will be questions it answers wrongly, or cannot answer, or on which it never stops: the correspondence formula always stops when it writes an answer [1], but the chains it sends to the calculating engines need not. But the same theorems say that any method that works by rule has such questions; they do not say that the seer is free of this limit, and no one has proved that he is. The White Tower Society laid down the rule that insight must be reckoned precisely because what insight yields is sometimes right and sometimes wrong, and counts only when it is checked. I said in § 12 of that paper that if a seer could see, for every rune chain, whether it stops, and were never wrong, his gift could not be written as rules. That still holds; but it is a conditional statement, and no one has yet shown that there is such a seer, who is never wrong. When an engine answers a question wrongly we feel that we have scored a point; yet we ourselves have answered many questions wrongly. In the fair-copy test the examiner can put only a few questions; he does not know which engine sits behind the curtain or what its weights are, and so cannot frame the one question that would defeat it alone; and the questions the engine cannot answer rightly are not necessarily answered rightly by the scholar.

I think that those who hold the mathematical objection would mostly be willing to discuss the matter on the ground of the fair-copy test. What they wish to say is that there are always questions an inference engine will not answer rightly, and with that I entirely agree. But we should not, because a machine has questions it answers wrongly, feel ourselves a grade above it. We too have questions we answer wrongly, and we never know which they are.

(4) The Argument from Knowing. An inference engine does not know what it is saying. When it cites a book wrongly it speaks as fluently as when it cites rightly; a scholar who truly understood would not do so. Forster of the Derola Medical College put it most clearly in a lecture of F.E. 358: “Not until an engine writes an argument because it has thought of it, and is glad to have thought of it, and would be ashamed to have it wrong, and not because the codes happen to have fallen into that order, can we say that it is the equal of a scholar, however well it writes.” [7]

On this view the only way to know whether an engine understands would be to be the engine, and to feel what it feels. By the same reasoning, the only way to know whether a scholar understands would be to be that scholar; and we should then believe that no one understands but ourselves. Few people speak in that way. In ordinary life we judge whether someone understands by how he answers and how he bears further questioning. So it is in the oral examinations of the Month of Scrolls: the examiner does not merely hear the apprentice recite an answer, but questions him further.

Those who hold the argument from knowing may put it more weakly: that performance in question and answer is only one piece of evidence of understanding, not sufficient evidence. That is fair; but then what we need is more evidence, not evidence of another kind: whether the engine can correct its own mistakes, say where it is unsure, and carry a principle over to questions it has not seen. That an engine speaks as fluently when it cites wrongly is just why it should be asked how sure it is. All this can be asked in the fair-copy test. Whether it has feelings of its own, question and answer cannot test; but the feelings of other men we have never tested directly either.

Here, for example, is a piece of further questioning in the fair-copy test.

Q: In The Copyist [8], if you were the author, would you move the page on which the copyist writes the Second Question in the margin to the very end of the book?

A: No. Where it stands, in the middle, the reader wonders at every later entry in the accounts whether he is reckoning even numbers again; moved to the end, everything before it is only a book of accounts.

Q: But at the end the reader would learn only when he had finished what the copyist had been thinking all along. Would that not be better?

A: Then it would be a riddle. Those who read that book like to follow his hand and think it out step by step with him, not to be told at the end.

If an engine could go on answering in this way, I think that most of those who hold the argument from knowing would no longer say that it is merely putting letters together.

(5) Arguments from Various Disabilities. An engine may perhaps do all that has been said; but there will always be something it cannot do. It cannot be kind, or make a jest, or enjoy the Twin Towers race, or tell new wine from old; it cannot learn from experience; it cannot make mistakes.

These claims rest mostly on an induction: the machines we have seen do none of these things, therefore no machine can. It is like an apprentice of Copyists' Street who has seen only one hand and believes that all letters everywhere are made so. Two of them deserve separate mention. Learning from experience is taken up in § 7. As for being unable to make mistakes, this has lately become the inference engine's best-known failing: it cites wrongly, reckons wrongly, and gives answers plausible in form and false in substance, and the rule laid down by the Academic Council about F.E. 355 was made for this very reason [9]. The old saying that machines never err is thus no argument, any more than the saying that they do nothing but err.

Two kinds of error must be distinguished here. One is an error of working: a crystal flickers, a cell of the store is written wrongly, and the machine does not do what its formula says. The other is an error of conclusion: the machine takes every step as its formula says, and what it concludes is false. Those who say that machines do not err are thinking of the first; the errors of inference engines are almost all of the second. The sum in the questions of § 2 is out by a hundred; the examiner

may ask himself whether a scholar reckoned it wrongly, or an engine reckoned it wrongly on purpose.

Among the disabilities there is one more: that a machine cannot be the subject of its own thought. This is not certain either. Among the documents of the repository that the first inference engine read was Lynn's own paper on the correspondence formula; ask it what the correspondence formula is, and it can tell you, and what it tells you of is its own kind. This shows only that there is no reason to forbid a machine to take its own make as a subject; whether it knows itself must be asked of its own version, of what it has just done, and of where it went wrong, and not of the books it has read.

(6) Ashby's Objection. Ashby said in the preface to Universal Rune Units that the universal units originate nothing, but can do whatever we know how to order them to do [10]. On this view an inference engine does only what it has been ordered to do and produces nothing new, and there can be no question of insight.

Ashby spoke of the universal units, and rightly; but the word "order" must be looked at closely. What a rune chain does is indeed fixed by its formula; but to be fixed by its formula is not to be known beforehand by the man who wrote the formula. That is just what Theorem 2 of my paper says of chains whose store may be extended without limit: even with the table of such a chain set out before us letter for letter, we cannot in general foretell whether it will stop [2]. An inference engine with a fixed store can, by Theorem 5 of the same paper, be judged in principle; but its states are so many that "in principle" and "in practice" lie very far apart. In either case the man who wrote the formula has in practice no way of knowing beforehand what it will do. The weights of an inference engine were written in by no one, and what it will do once it has learned is still less known beforehand to those who taught it. In Royce's theorem of the twin primes of F.E. 357, the set of sieve weights that the engine chose out of some hundreds of thousands of candidates was one, Royce himself said, that he had not expected [11]. The set was new to Royce and was not in the repository; but it was Royce, not the engine, who proved that it worked. The example answers only the claim that whatever a machine does was foreseen by men; it does not show insight. Inference engines surprise me often; when we say that a thing produces nothing new, it is often only because we have not tried to foresee it.

There is a view akin to Ashby's objection that may be taken with it. A thought sent into most minds draws out one or two thoughts in reply, as a share of mana sent into an array below its threshold is answered once or twice and dies away. But there are a few minds that have passed the threshold: a thought sent in draws out a whole train of thoughts, as a share of mana sent into an array above it calls forth ever more answering mana. We might put our question so: can a machine be made of the kind that has passed the threshold? This is only a figure; what passing the threshold would mean in an inference engine, I cannot yet give a test for.

(7) The Argument from Continuity. The mana of living beings flows continuously, and insight comes with it; a rune chain goes step by step and cell by cell, and is discontinuous. What is discontinuous cannot learn what is continuous.

Since F.E. 338 this has not been easy to say. Selby and Ormes have shown that the exchange of mana between the Web of Laws and the depths of Tiamat proceeds wisp by wisp [12]: the continuity of mana was an approximation of the classical theory. The wisp shows only that the exchange of mana goes by shares, not that everything in a living being is discontinuous; I use it only to show that the continuity of mana cannot be taken as a premiss. Even apart from this, in the fair-copy test a discontinuous machine can imitate what is continuous closely enough: a star-reckoning array unfolds its tables step by step, yet the tide-tables it computes agree with the continuous rise and fall of the tide, and no reader of the tables can see where the steps are. The Tower of All Things had also, of old, a tide-disc, which reckoned the height of the tide by the flow of water and was a continuous instrument. If an inference engine were asked in the fair-copy test what the tide-disc would answer, it need not answer exactly; provided the disc's initial state and constants can be reckoned, the time and precision observed are limited, and the course of the water over that time has an error that can be bounded by reckoning, it need only give the several answers the disc might give (for the initial state is never measured without some error), each with its proper chance and with the proper connexion between one answer and the next, and the examiner could not tell it from the disc.

(8) The Argument from Informality of Behaviour. A man does not act by rules: no set of rules can be written that says how a man will act in every circumstance. A machine acts by rules. Therefore a man is not a machine.

Two things are confused here: rules of conduct, such as "stop when the alarm sounds", and the laws of behaviour, that is, whatever laws a man's actions in every circumstance obey. That a man has no set of rules of conduct that could be written out in full is true; that his behaviour obeys no laws at all we have no evidence. That we cannot find such laws does not mean that there are none.

I have set up on a calculating engine of the Kanaich Institute of Technology a small chain using only eight hundred cells of store: given a number of twelve figures, it replies within two seconds with another of twelve figures. I have not yet met anyone who could learn from these replies the rule by which they are made. This shows only that not seeing a rule is not the same as there being none; that there are laws, that the laws can be reckoned by a chain, and that we can find them, are three different things.

(9) The Argument from Perception. A caster can feel the tide of mana, and a seer within the White Tower can feel the laws; an inference engine feels nothing. Worse: in the fair-copy test, a scholar behind the curtain who knew a perceiving string-spell could learn what was written on the slips in the examiner's hand, and the engine could not.

This I think the hardest objection to answer. The casters' common feeling is a fact; perceiving string-spells are a fact too, and string-spells are not suppressed by wardings against magic [13], so that an ordinary curtain or wall may not stop them; how far they reach and what does stop them this paper does not consider, and to try them would need leave under the law. If anyone cast such a spell

during the test, the test would be unfair. Fortunately every casting leaves its trace. That no one under examination may cast, and no one outside help, is an old rule of the examination hall. Before the test, the traces of the calculating engines and recording instruments working as usual are taken down as a base, and the rooms are searched for any lasting spell of perception cast earlier. The traces are examined by an independent warden, and only after the examiner has given his verdict, lest they betray in which room the engine sits; the signature of a trace shows its source, and its content is examined apart. Star-gold is rare, and serves only for the cases of primary standards and secondary standards of the first grade; an examination hall cannot be shielded with it. The test is held far from places of casting, out of reach of ordinary interference, and the tide and the supply are watched from a station, the sitting being stopped if they rise and fall too much. Against perceiving string-spells the safeguard is the examination of the traces after the test. No hall keeps out the tide, so the test had better not be held in a tide year. With these precautions, only the first half of the argument from perception remains: the engine cannot feel the tide. That is true; and the questions of the fair-copy test do not ask about things that must be perceived as mana, an exclusion which confines the test to such insight as can be written.

7. The Apprentice Engine

It remains to ask how an inference engine that passes the fair-copy test might be obtained.

The first inference engine was read into being: it read the code repository and was corrected whenever it read wrongly, as though an apprentice had read through the whole of the Great Library. But reading is not insight, and however many books are read, someone must still examine and correct. Before an apprentice comes out of his apprenticeship, his master must watch him write, correct what he writes wrongly, and tell him what he writes rightly.

The mind of a scholar may be thought of as made of three things: what he was at birth, the teaching he has received, and the other experience he has met with. An inference engine may be thought of in the same three parts: its first weights, the teaching of checking, and the books it has read.

I think that an inference engine might be taught in the same way, and taught by the White Tower's own rule. Each argument or construction it proposes would be sent to be checked; those that pass would be marked right, those found in error marked wrong, and those that can be neither proved nor tried marked neither, and its weights adjusted accordingly. Where to begin on a problem, and judgments of doctrines and of verse, can be neither proved nor tried; for these it would be taught as a master of the Copyists' Street corrects an apprentice's fair copy, scholars marking whether its reasoning holds and its instances fit, and its weights adjusted by their marks, though nothing is recorded as right or wrong. To teach so is to seek more results that can be checked, not to settle every truth; and where casting or construction is concerned, the rules of safety and of leave must be kept. That is just what the rule that insight must be reckoned means, applied to a machine. What an

engine so taught will do once it has learned will not be wholly known to those who taught it; and that is another answer to Ashby's objection.

It may seem a contradiction that a machine acting by rules should learn to change its own rules. It is not strange. In a trained inference engine part of the formula does not change, how correspondence weighs, for instance; another part does, namely the weights, which change as it is taught. So it is with men: the rules by which we read and write change all the while we are learning, but the rules of learning do not.

In teaching it, it may be well to let it draw lots at certain points: where several paths all seem reasonable, to take one at random. An apprentice who walks only the paths his master has walked will never walk out of his master's paths.

We sometimes look at a man's mind as we might take apart one of the bound books of Chaen. We lift off the cover, then the endpapers and the label, and at each layer we may say, "This is only the binding, not the book." When everything has been lifted off, is there only a stack of blank paper left, or is there another layer, the true book? If there is only blank paper, then the whole book was binding; and on that view, the binding a machine makes need not be other than a book.

To teach an inference engine in this way will probably take many years, the spare capacity of many engine rooms, and many people willing to spend their time in checking. We live long, and this is just the work for us. We cannot see far ahead; but the stretch that can be seen is enough to keep us walking for a good while.

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